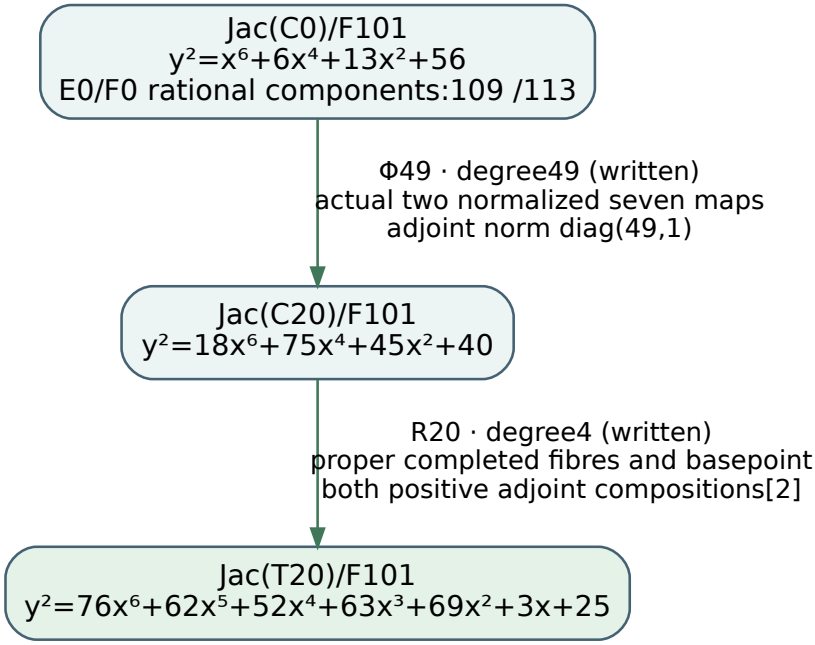
Proper C20→T20 map and the degree196 chain

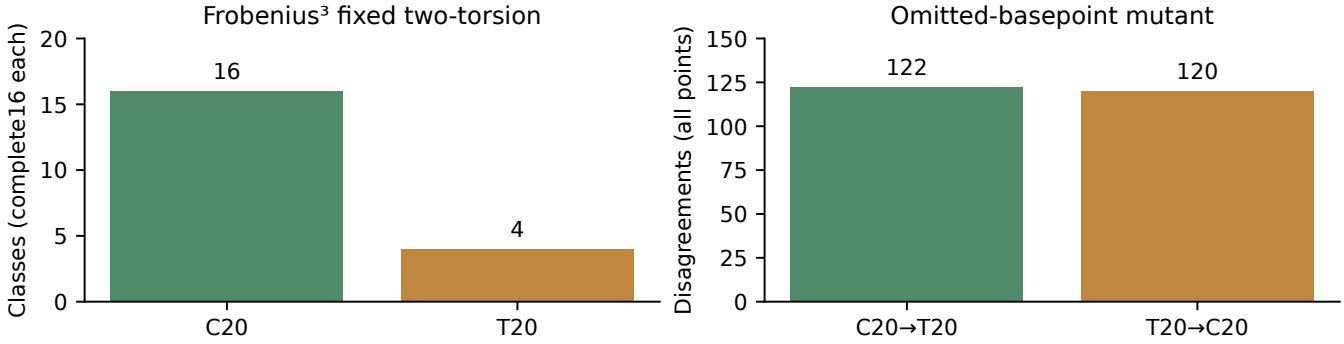
2026-11-12 · full41 /6cecba · point-selected02 /19c509 · unsigned

All12317 rational C20 Mumford inputs map to distinct valid Frobenius-fixed T20 classes;both positive dual compositions equal actual doubling. Public6159 recovery succeeds on every input. Actual C0→C20→T20 composition is a bijection on all12317 classes,with adjoint norm diag(98,2),recovery and all109+113 scalar images/exact orders. Scientific standing remains unsigned. [pcc:unverified]



Complete41 /6cecba · all12317 rational classes
C20→T20 and C0→C20→T20 bijections
Chain degree196; adjoint norm diag(98,2)
Actual recovery and all109+113 scalar images
Unsigned; coordinate-key caches, no DLP labels

Global degrees and polarizations are written Picard/AV results;the complete native run checks actual finite classes. It uses physical Mumford keys for caches,never source enumeration indices or logarithm labels. [pcc:unverified]



Exact finite populations from full04/6cecba and point02/19c509,no sampling uncertainty. Source122 point controls include both geometric infinities of the nonsplit source;they are not122 F101-rational curve points. [pcc:unverified]

Complete41 tests retain all old39,with zero failed/ignored/filtered;old reference+legacy guards,fmt and strictClippy pass. All627 source hashes are unchanged. Both geometry6141-byte and kernel24475-byte JSON outputs are byte-identical to their accepted originals. Whole155310-byte log is standardXZ12028,with raw/archiveSHA256 and exact roundtrip. [pcc:unverified]

Proper fibres, formal boundary and retained failures

For D=ΣP_i−dW,use R(D)=ΣA(P_i)−dA(W),where A(P) is the completed degree-two target fibre relative to2W′. Finite fibres retain full U,V polynomials;degree-drop keeps the finite point and its signed infinity. Both source infinity signs,all six branch points and chart corners are included. The negative swapped correspondence fixes the x−z sign to give the positive adjoint. Written proof720a59 details the limits,Picard map, prime-field descent,degree196 and nonuniform recovery. [pcc:unverified]

Point-selected01/bf3f68 passes1/40filtered with all627 hashes;its observed repeated target-fibre count is0. Before whole acceptance,the test-only coverage addition preserves that positive run and adds actual repeated Mumford input doubles122/120 and all256 two-torsion map additions. Static02/abe0b7 and selected02/19c509 pass. Forced degree-drop inputs are4 each direction;both signed infinities and every branch case execute. No target repeated-fibre coverage is inferred from repeated source inputs. [pcc:unverified]

All16 actual geometric two-torsion classes are checked in each direction, including all256 group sums and map additions,kernel4/image4 and actual matching kernels. Frobenius³ fixes16 source classes and4 target classes. The canonical six-branch pair model and the Galois-equivariant restriction of any odd-degree isogeny then exclude a direct odd-degree F101 isogeny between these two models. This independently applies the external CT49 parity observation;the degree4 bridge respects it. [pcc:unverified]

The wrong uncorrected transpose agrees with the positive norm only at identity:12316 disagreements,one equality. New exact bound local Lean passes d01c21/2de44b/507bb7/f2b3ce/87bce9/b7ad54 cover abstract order-based recovery/sign,explicit constants and canonical Fin6 pair permutations. Gate03 audits6/0trusted;project198written39modules. Actual curves,order, Picard correspondence,polarization,root relabeling and Weierstrass model remain explicit native/written inputs.224compiled is a new-CI expectation. [pcc:unverified]

The complete F101 class domain uses anchors in $F101^3$/$F101^6$ and support in $F101^{12}$. Forced x∈$F101^6$ point controls have Frobenius⁶ fixed-or-negated classes;their image u∈$F101^6$ is asserted and checked. Arbitrary $F101^{12}$ Jacobian inputs can need degree24 support and are outside this domain. The raw curves retain their original leading coefficients18 and76. [pcc:unverified]

Formal01/45889b and02/7fc8a6 fail with0audited;compiler recovery sorryAx is rejected. Explicit module identities and reflexivity fix03 without changing statements or hypotheses. Additive76b1b9 corrects a placeholder in failed02 metadata. Whole-selected03/f776e2 prints pass but lacks its original exit/finish/afterhash receipts:preserved incomplete,not accepted. Its archive caller failed to stop;full04 was already launched and then completed all original criteria. The separate failure report retains each attempt and its evidence. No exit is fabricated from test stdout. [pcc:unverified]

LocalLinuxx86_64 correctness/development,Rust1.99.1/Lean4.35.0,no cost measurement or benchmark isolation. Existing algebra23/10certificates remain unchanged. Root30 CI43283121414 succeeds25/38/23/eight/214compiled/ 188written37modules0trusted (965785);c7 own43291074099 is pending at the older39-test scope,next20:24UTC. Consumer1513 pins338d35 and its three-job CI is settled success3ea63d. New proper41 replay is next. [pcc:unverified]

Sources:BD0902.3480§4;nativeRic+legacyCantor/Chart;720a59/b24be8/128eb1/ afd164/063773;frozen a2c024/b0c5f6;external CT49f7d4c6be. KB unavailable, actual fallback receipts retained. REPORT.md links complete custody, sources,editable graphs/data/renderer and the inspected final PDF. [pcc:unverified]